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On-the-Fly SfMv2: Get Better from What You Capture

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Abstract—**In the last dozen years, Structure from Motion (SfM) has been a constant research hotspot in the fields of photogrammetry, computer vision, robotics etc., whereas, real-time performance is widely concerned recently. In this work, based on the original on-the-fly SfM which runs online SfM while image capturing, we present an updated version (on-the-fly SfMv2) with three new improvements to get better from what you capture: first, online image matching is further boosted by employing the Hierarchical Navigable Small World (HNSW) graphs, thus more true positive overlapping image candidates are faster identified; second, a reasonable self-adaptive weighting strategy is proposed for hierarchical local bundle adjustment, which is expected to improve the SfM results; finally, but most notably, we incorporate with the capability to deal with images from multiple agents, and multiple reconstructions are seamlessly merged into a complete model when commonly registered images appear. The comprehensive experiments demonstrate that our on-the-fly SfMv2 is able to generate better results from what is captured, in particular, a more complete reconstruction is generated in a higher time efficient way, yet state-of-the-art SfM results is obtained.**

Index Terms—**Structure from Motion, Real Time, Overlapping Image Pairs, Bundle Adjustment, Multiple Agents**

I. INTRODUCTION

In recent decades, SfM has increasingly gained attentions by the researchers from various communities and numerous endeavors have been spent on this study. As a consequence, abundant awesome achievements have been made, which can be mainly divided into three categories: Incremental SfM [1,41,52,56], solving new images sequentially; Hierarchical SfM [12,28,50], clustering images into overlapping subsets, which are then oriented hierarchically; Global SfM [8,22,53,54,55], estimating poses of all images synchronously. However, due to the intensive computations required by some inherent procedures (feature extraction and matching, two-view geometry verification, image pose estimation using perspective-n-point, triangulation, bundle adjustment etc.), the vast majority of SfM methods work in an offline mode, namely, all images are first collected and then input together into a specific SfM pipeline to estimate image poses and corresponding sparse point cloud. While these methods have shown their merits regarding accuracy and robustness, the offline processing limits real-time applications, such as online measurements and quick quality assessments, etc.

To cope with the real-time requirement of estimating poses and 3D map points, another hot research topic – VSLAM (Visual Simultaneous Localization and Mapping) emerges, which takes sequential frames as input to output the camera trajectories and 3D maps in real time. In general, based on different embedded sensors, there are mono-VSLAM, stereo-VSAM and inertial-VSLAM [5,33,34]; for different tracking methods, there are feature-based VSLAM [34] and direct VSLAM [10]; with different optimization methods, we have Kalman-based [9] and graph-based VSLAM [16]. All these VSLAM methods constitute several common parallel processing threads that deal with modules of tracking, map generation and optimization, loop closure detection. One implicit assumption of VSLAM requires that the input data must be video or sequential frames, i.e., frames must be spatiotemporally continuous and two adjacent frames should be continuous in time and space. In addition, to mitigate the limitation of error accumulation, loop closures are frequently employed in most VSLAM methods to correct trajectory and improve maps after a long-term tracking, which leads to the revisitation of previously mapped areas during data collection. As a consequence, the data collected for VSLAM is hindered by the constraint of spatiotemporal continuity (better with loop closure).

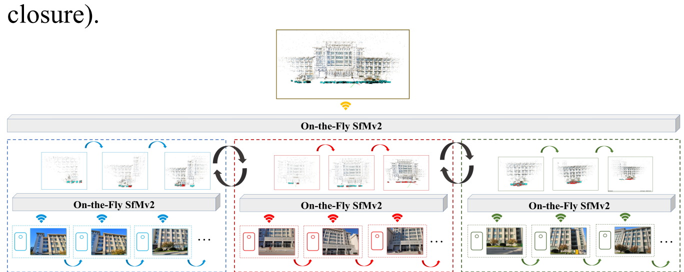

Fig. 1. Overview of our proposed on-the-fly SfMv2.

In this work, we build on our previous on-the-fly SfM [63], a new real-time system able to run online SfM while image capturing, achieving the goal of yielding image poses and 3D sparse points (as traditional offline SfM typically outputs) in an online mode (similar to how VSLAM works) by images captured in an arbitrary way. Here, as Fig.1 illustrates, we go one step further to explore the possibility of **handling multiple agents**, **multi-reconstruction association and merging**, presenting an updated version of on-the-fly SfMv2 to get better from what you capture. The main improved contributions are:

- **Improved online image retrieval**. Instead of using a

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pre-trained vocabulary tree, we now employ the so-called Hierarchical Navigable Small World (HNSW) [30] graphs for improving the online image retrieval. HNSW graphs are superior in retrieval precision and time efficiency, and can be dynamically built with new fly-in image, which is more tailored to the proposed on-the-fly SfMv2;

- **Self-adaptive weighting for local bundle adjustment (BA)**. To further boost the performance of local bundle adjustment, a self-adaptive weighting strategy for hierarchical local BA is adopted in the new presently system. Unlike the first version [63] that investigates an empirical setting of power function for weights, this work takes profit of the similarity degrees generated in the image retrieval stage to estimate more reasonable weights self-adaptively.
- **Capability for multiple agents and reconstructions**. The most notable update of our new On-the-Fly SfMv2 is the capability to run online SfM with multiple agents capturing images, and to associate and merge various reconstructions. Thus, better results with complete reconstruction can be efficiently obtained. This is achieved by the new interface that allows receiving data from multiple cameras, and the fact that different reconstructions are associated by the commonly registered images and merged seamlessly based on the corresponding 3D similarity transformation.

II. Related Works

In the past decades, ample works have been studied on SfM, among which some focus on improving the time efficiency and achieving real-time performance. In this part, we will review four related topics including SfM, VSLAM, image retrieval and efficient bundle adjustment.

A. SfM (Structure from Motion) and VSLAM (Visual Simultaneous Localization and Mapping)

Both SfM and VSLAM focus on generating geometric information from data of images, more specifically, they take images as input and accurately output the image's poses when exposuring and the observed 3D object points' positions, which are basically consistent to expected output of this work.

SfM. SfM can be mainly divided into three categories: incremental SfM, hierarchical SfM and global SfM. Incremental SfM, solving images sequentially with iterative bundle adjustment, has been widely explored, whereby many popular academic packages are successfully developed. Bundler [45] is one of earliest open resources that deals with large-scale unordered image collection, to visualize the intermediate reconstruction results during incremental processing, VisualSFM is implemented by [57] with friendly GUI (Graphic User Interface). Recently, three packages, namely Theia [47], OpenMVG [32] and Colmap [41], frequently appear in relevant works, especially, Colmap has obtained wide attention because of its superiority in stability, robustness and high accuracy. Thanks to these kind open-sourcing frameworks, many endeavors have been done to improve incremental SfM. Wu [56] adopted an iterative re-triangulation strategy to cope with error accumulation resulted from noisy poses. Schönberger and Frahm [41] made three main contributions in Colmap, two-view geometry is verified by considering essential, fundamental and homograph matrix, newly added image is selected based on the number of visible tie points and their distribution on each candidate image, ransac-based DLT is applied to triangulations. Wang et al. [52] solved exterior rotation and translation separately, rotation is robustly estimated using relative rotations between new image and registered images and translation is obtained by a linear equation system. Nearly the same time when incremental SfM emerged, hierarchical SfM started to be researched, whose general idea is to split all images into overlapping sub-clusters and hierarchically merge each sub-cluster's SfM results. Time efficiency is improved due to the easy parallelization of each sub-clusters. Zhao et al. [64] proposed Linear SFM which begins with small local sub-cluster using nonlinear bundle adjustment, the solved sub-clusters are hierarchically merged by using a linear least square and a nonlinear transform. Michelini and Mayer [31] started from triplets and hierarchically merged connected triplets, and time efficiency was further improved by randomly ignoring shared tie points. Liang et al. [27] proposed a hierarchical SfM tailored for large-scale oblique images, they applied GPS/IMU and terrain information to group images effectively which shows good accuracy and time efficiency. To further speed up the processing of SfM, the aforesaid iterative and hierarchical solution is gradually updated by global SfM, as global SfM handle all images simultaneously for estimating their corresponding initial external orientation parameters and only one final bundle adjustment is run for optimization. To solve global rotations, many researches take relative rotations (from two-view epipolar geometry) as input and estimation rotations for every image that can minimize the error distance from corresponding relative rotations, Chatterjee and Govindu [6] first applied L1 norm rotation averaging based on lie algebraic, then the results were refined by a iterative re-weighted least squares with Huber-like loss function, this method was claimed to be more computationally efficient and robust than Crandall et al. [7] and Hartley et al. [17]. Based on the estimated global rotations, global translations of each image can be solved. Wilson and Snavely [55] presented a non-linear optimization system which investigates the direction differences of relative translation and global translation. Zhuang et al. [68] explored the various lengths of baselines and found that the accuracy of global translations is more sensitive to longer baselines, they suggested an angular error function that works well on various baselines. Wang et al. [53] presented a linear equation system to first compute globally consistent scale factors for relative translations, and global translations are then simultaneously estimated by these scaled relative translations. In the next two years, Wang et al. [54] investigated the geometric properties of essential matrix for multiple views and solved estimated global rotation and translation synchronously if the corresponding images are not collinear.

VSLAM. To cope with real-time performance that is one of the most significant concerns relevant to this work, there exist many well-developed VSLAM methods that are capable of online tracking and mapping and should be reviewed here. Presently, VSLAM is typically developed in two main directions: first, Feature-based methods, such as PTAM [24], VINS-Fusion [15], PL-SLAM [36] and ORB-SLAM series [5,33,34]. These methods focus on extracting and tracking distinctive features in the environment to estimate the pose of each frame and build a map. Generally, keypoints, corners or lines are extracted as features, which are then matched across successive frames to track motion state and reconstruct structure. Such methods have been widely and successfully tested on various environments (e.g., indoor benchmark [46] and outdoor benchmark [14]). However, they might degenerate in the case of poor texture and repetitive patterns; second, Direct methods. To ensure a long-term tracking in complex scenario, especially in texture poor places, direct methods, such as LSD-SLAM [10], SVO [13] and DSO [11], estimating poses and dense (or semi-dense) structure, were proposed according to the intensity information among neighboring frames. The inherent assumption is that the 3D objects scene's appearance on successive images keeps constant over very short periods, this can result in effective operation in low-texture environments where feature-based methods might fail. However, these direct methods are sensitive to lighting changes and require good initializations. On the other hand, learning-based methods are also integrated into various modules of VSLAM. For example, in the front end, DXSLAM [26] replaced the original hand-crafted features with deep features extracted by CNNs and trained a new corresponding Bag-of-Word structure, similarly, Bruno et al. [4] applied the learned local feature - LIFT [60] as input observations. In the back end, Tateno et al. [49] proposed CNN-SLAM and the dense depth maps were predicted by CNNs with sparse depths from monocular SLAM, Tang and Tan [48] presented a tractable network for bundle adjustment and several basis depth maps were first predicted which are then optimized via a linear combination to yield final depth.

Unlike the conventional SfM methods that input all collected images together and conduct subsequent geometric processing, this work implements an online SfM framework while multiple agent-based image capturing without professional photogrammetric regulations. And, in contrast to VSLAM, the necessity of spatiotemporal continuity is not required anymore, nor is the pre-knowledge of GPS/IMU. Two most relevant works are the so-called real-time SfM [65] and our previous on-the-fly SfM [63]: the former proposed a hierarchical solution to improve image matching using BoW and multi-view homography, but the spatiotemporal continuity between images is still required; the latter work is initial version of our On-the-fly SfMv2, whose improvements are explained in **Section I**.

Note that while there are ample excellent SfM and VSLAM methods worth reviewing, we only select a few popular and related works to review.

B. Efficient Image Retrieval

Image retrieval has been playing a crucial role in SfM and VSLAM, as it is widely used to fast identify overlapping image pairs and detect loop closure. As for online SfM, the first step is to find the correlations between new added image and already registered images, therefore, efficient image retrieval methods are even more crucial under this context. Nowadays, relevant studies are revolved around two directions: feature refinement and efficient indexing structure.

Features refinements. Image feature is often used to calculate corresponding similarity degrees, which can determine how precise the retrieval performance is. Refinement of local features: local features are typically with high number and high dimensional descriptor, which leads to intensive computation for finding nearest neighbors. To reduce the number of features, Wu [56] found SIFT features of higher scale are more possible to be correspondences and then only considered to features with higher scales for fast matching, it is called "preemptive matching". Hartmann et al. [18] trained a classifier of random decision forest that input SIFT feature descriptor for predicting its matchable probability, and image matching is accelerated by using less matchable features. Michelini and Mayer [31] map the original SIFT feature from 128 real space to {0,1} hamming space, thus the similarity degree can be easily estimated using Streaming SIMD Extension. Refinement of global features: global features are now more popular on retrieving similar images, mainly due to its high efficient computation and the superiority of learning-based global features. Razavian et al. [38] proposed a simple and efficient method to extract global feature from off-the-shelf CNN model, max pooling operation was applied on the last activation map channel-wisely. Arandjelović et al. [3] changed the last pooling layer of conventional CNN models with a trainable pooling layer via a soft assignment for VLAD, which enhance the performance the place recognition from images. Radenović et al. [37] first proposed an automatic annotation method for generating similar and non-similar image pairs by exploiting SfM result, based on which pre-trained CNNs was then fine-tuned for better global image features. Shen et al. [42] generated ground truth overlapping image pairs via exploring the 3D mesh model and the reprojected triangles from mesh to images were investigated on pairwise images, CNNs were then adjusted to make extracted global features more sensitive to matchable pairs. Ren et al. [40] employed GNN (Graph Neural Network) to further prune the similar candidate images from CNN-based global features. Recently, Hou et al. [21] proposed a CNN fine-tuning method integrating multiple NetVLADs to aggregate feature maps of various channels and published an benchmarks LOIP that consists of both crowdsourced and photogrammetric images.

Efficient indexing structure. After extracting features from images, indexing structures are typically built to efficiently retrieve the nearest neighbors for target query. Bag-of-word [43] is a pre-trained hierarchical indexing tree with cluster centers as nodes, based on tf-idf method, each image is represented as a vector to efficiently find similar candidates. VocMatch [19] used SIFT features to train a large two-layer vocabulary tree with 4096 nodes in the first layer and 4096×4096 nodes in the second layer, correspondences should be localized in the same node. Zhan et al. [61] improved Bag-of-word by presenting multiple hierarchical indexing trees and GPU was used to further improve the retrieval speed. Later, instead of using local image features, CNN feature maps from a pretrained VGG-16 network [44] were used to construct a bag of word (BoW) model by Wan et al. [51], and image's representing vector is formed by the CNN-based BoW. Wang et al. [53] suggested a random KD-forest to build multiple self-independent KD-trees, which increase the possibility to find real nearest candidates without the time-consuming back traversing procedure. Jiang and Jiang [23] studied the statistical information of the similarity scores estimated via vocabulary tree, which were shown to be able to select more image pairs by expanding minimum spanning trees (MST) and avoid the photogrammetric block breaking apart. Inspired by VocMatch, Zhan et al. [63] unsupervisedly trained a hierarchical vocabulary tree using global features [21] and similar images' global feature are expected to be classified into the same node, this method has been successfully embedded in our previous on-the-fly SfM.

C. Efficient Optimization of Bundle Adjustment

Thanks to the developments of relevant theories during the last few decades, bundle adjustment (BA) has been widely studied and has now become a well-developed technique for optimizing image poses and 3D point positions, yet it is still one of most time-consuming steps among all geometric processing, this is particularly crucial for online real-time SfM. As the number of enrolled images increases, quite a few methods for solving BA in a time efficient and stable way are developed. For example, Agarwal et al. [2] found that large-scale BA typically leads to a large linear equation system and the canonical Schur complement was still not fast enough, they then explored preconditioned conjugate gradients (PCG) to iteratively yet efficiently estimate the inverse of large positive definite square matrices. Based on PCG, Wu et al. [58] proposed multicore bundle adjustment and further improved the speed of BA by implementing PCG on Multiple CPUs (Central Processing Unit) and GPU (Graphic Processing Unit). Zheng et al. [66] extended the application of multicore bundle adjustment on extremely large high-resolution UAV images and demonstrated its superiorities. Huang et al. [20] presented DeepLM, based on backward Jacobian Network, they developed a general and efficient Levenberg-Marquardt (LM) solver that can automatically compute sparse jacobian matrix. To cope with large-scale BA problem, distributed methods that partition large BA problem into several overlapping small subset BA attract researchers' interests. Zhang et al. [62] solved each subset BA in parallel, with the global camera consensus constraint, all refined subsets are merged into one complete block iteratively using ADMM (Alternating Direction Method of Multiplier) algorithm, Mayer [29] took shared 3D tie points as global consensus constraints and better convergence was achieved by integrating the corresponding covariance information. MegBA [40] parallelly solved subsets via multiple GPUs, which provide a more time efficient solution. Recently, Zheng et al. [67] proposed DBA (Distributed bundle adjustment) with block-based sparse matrix compression, and LM method can be exactly applied on the reduce camera system of divided subset.

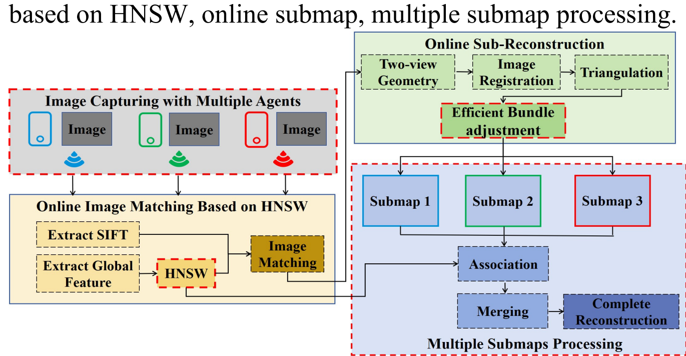
All the above reviewed works try to approach an efficient BA optimization by taking all unknowns (including poses and 3D points) simultaneously, this is inherently not feasible for incremental, sequential and online SfM mode, as it is obviously not a wise choice to run global BA after each and every new image is added. To address this issue, VSLAM approaches [5,34] applied a sliding window to limit the number of images that are enrolled in BA, which resulted in a small optimization problem on several neighboring frames (in temporal and spatial space). This simplification effectively reduces global BA to local BA, trading off precision for saving processing time. Colmap presented a strategy of alternating between local and global bundle adjustments to compromise precision and processing time. Nonetheless, the fixed number of images involved in the local adjustment may not cover all association images, which might affect the adjustment's accuracy.

III. ON-THE-FLY SFMV2

In this section, more detailed improvements of the updated on-the-fly SfMv2 are explained. First, we show the overall workflow of our new on-the-fly SfMv2. Then, three improved contributions of this system are introduced: 1) Faster image retrieval using learning-based global feature and HNSW; 2) Improved efficient local BA optimization via self-adaptively weighted hierarchical tree; 3) Submaps [1] association and merging.

A. Overview of On-the-Fly SfMv2

Fig.2 shows the general workflow and main components of our new system, that are basically analogous to our previous on-the-fly SfM with some new significant characteristics highlighted by red dashed boxes. It mainly includes four parts: image capturing with multiple agents, online image matching based on HNSW, online submap, multiple submap processing.



[1] In the whole paper, we do not distinguish between the concepts of submap and sub-reconstruction.

Fig. 2. Example of the proposed on-the-fly SfMv2 workflow. Three mobile individual agents are shown, the corresponding new updates are highlighted by the boxes with red dashed boundaries.

Image capturing with multiple agents. Unlike our previous on-the-fly SfM that can only deal with single agent and wireless WIFI transmitter is applied for image transmission, on-the-fly SfMv2 makes following updates to simplify and speed up image collection: first, on-the-fly SfMv2 is compatible with various platforms, such as mobile phones and iPad, which can improve the flexibility of our work; second, multiple agents working mode is supported to deal with images from various sensors, which is expected to improve the time efficiency. Third, real-time image transmission of on-the-fly SfMv2 is achieved with local area network, 4G and 5G, which enhances the practicability of the system.

Online image matching based on HNSW. For online real-time SfM, the accuracy and time efficiency of determining overlapping image pairs for newly fly-in images are very vital factors for subsequential geometric processing, moreover, multiple agents require faster image retrieval due to more images are flown in. In the new on-the-fly SfMv2, the learning-based global features [21] are retained, but the method of HNSW [30] is used to determine the matching relationship between new images and registered ones, so as to ensure the speed and accuracy of online image matching. (More details can be found in **Section III.B**)

Online submap. This part mainly focuses on the estimation of image pose and 3D point, which is basically identical with on-the-fly SfM. In particular, two-view geometries are first verified using various models (essential matrix, fundamental matrix and homography matrix) [41] and an initial stereo reconstruction is selected, then, EPnP [25] and RANSAC-based multi-view triangulation [41] are used to solve image registration and triangulation problems. In this work, the way that how newly fly-in image should affect its connected overlapping images is studied, we propose a local bundle adjustment with self-adaptive hierarchical weights to robustly and quickly solve the most time-consuming bundle adjustment (See **Section III.C** for details).

Multiple Submap processing. One of the most notable improvements of our new on-the-fly SfMv2 is the capability to deal with a set of submaps, which in practice frequently happens (especially in the case of capturing images arbitrarily), for example, one single agent takes images on one building façade and then directly visit another façade for collecting images until overlapping images emerges, or two agents take images of the building from different facades and gradually overlap with each other. Therefore, in on-the-fly SfMv2, after finishing submaps, their associations are firstly fast identified via the proposed HNSW-based image retrieval method, and then, associated submaps are merged into a complete reconstruction by a robust and efficient merging solution (See **Section III.D** for more details).

B. Faster Image Retrieval Based on Learning-based Global Feature and HNSW

For the updated on-the-fly SfMv2, to ensure online image matching for multiple agents and fast association identification of submaps, we investigate a faster image retrieval method by Hierarchical Navigable Small World (HNSW) graphs. Fig.3 presents the workflow of our faster image retrieval and the key modules: 1. Pre-train models. A pre-trained CNN model is selected as global feature extractor [3,21,39]; 2. Fast image retrieval for fly-in image. For one new image, pre-trained CNN model is firstly used to extract global feature, which is then input into HNSW to incrementally refine the corresponding indexing structure and to dynamically fast identify matchable candidate images.

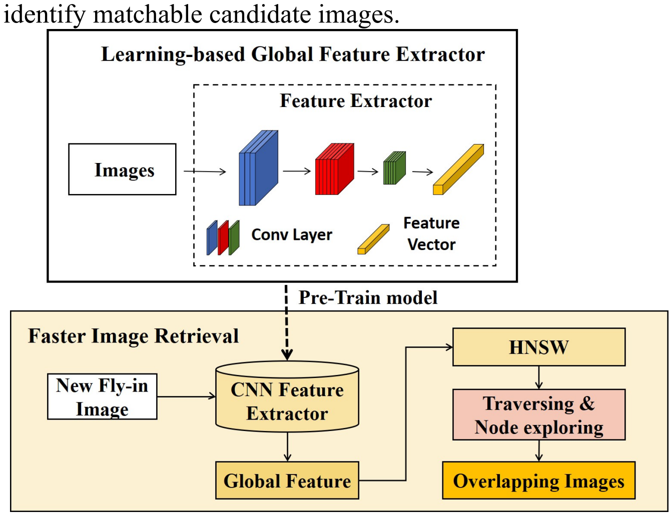


Fig. 3. Fast image retrieval workflow based on learning-based global feature and HNSW.

Learning-based Global Feature Extractor. CNNs have shown superiority on retrieving visual similar images as global feature extractor [46]. In this work, we still keep using the fine-tuned CNN model from [21] as our global feature extractor, as it is found that the model from [46] is technically explored for identifying overlapping image pairs, which is original for accelerating offline SfM and should be also suitable for the online image retrieval, and our previous on-the-fly SfM has demonstrated the efficacy of [46] for identifying overlapping image pairs.

Incremental establishment of HNSW. To the best of our knowledge, vocabulary tree (or its relevant variants) is one of the most common and effective methods for accelerating image retrieval in large-scale image orientation problems [19]. However, the time efficiency and accuracy of vocabulary tree-based retrieval heavily rely on the pre-training procedure, this is typically a very time-consuming task and may not generalize well to other unseen scenes. In contrast, HNSW can be incrementally constructed in real-time as newly captured image come in and pre-training is not needed anymore, which just fits very well to our on-the-fly SfM. The incremental establishment process of HNSW is illustrated in Fig.4. For a newly fly-in image, the global feature of the image is first extracted, and then the global feature is added to the HNSW structure as an inserting node. The number of layer that this inserting node start to reside can be calculated by the (1):

$$l = -\ln(unif(0,1)) \cdot m_L \quad (1)$$

where l is the new node's layers, m_L is a normalization factor for layer generation, and one simple choice for the m_L is $1/\ln(Max)$, Max is a pre-set parameters that indicates the maximum number of connections from one node to all the other nodes in HNSW. In general, for the layers above l, the node closest to the inserting node within each layer is identified as the enter point of the next layer, as the layer l and below, Max nodes closest to the inserting node are searched and their connections with the inserting node are added into HNSW graphs, Max is increased by two times in the lowest layer to ensure good retrieval recall (see more details in **Algorithm 1** or [30]). As more images are captured and fly in, the structure of HNSW graphs is constantly updated.

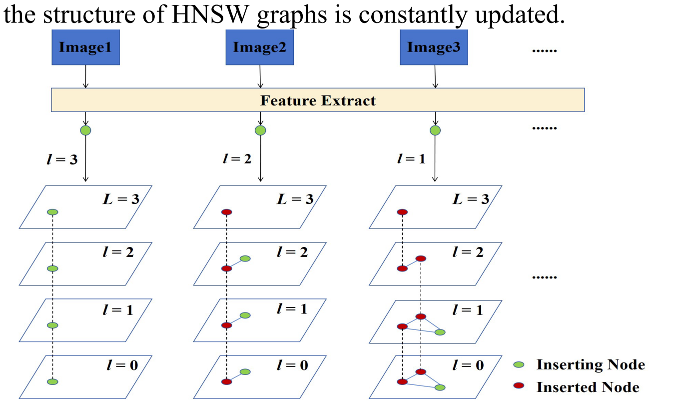


Fig. 4. The incremental process of HNSW establishment.

Algorithm 1 Establishment of HNSW

Input HNSW graphs, inserting node q, maximum connections for each node Max, normalization factor for layer generation m_L, size of dynamic candidate list *efcon*.
Output Updated HNSW graphs with inserting node q
1. **Symbols** W: list for currently found nearest nodes, *ep*: enter point, L:top layer, l: $-\ln(unif(0,1)) \cdot m_L$, layer of q.
2. **for** $li \in (L, l+1)$
 W = Search_layer(q, *ep*, *efcon*=1,*li*)
 ep = closest nodes from W to q
3 **for** $li \in (l,0)$
 W = Search_layer(q, *ep*, *efcon*,*li*) (search *efcon* nearest nodes at layer *li*, see more details in **Algorithm 2**)
 Select M closest candidate nodes from W, indicated as *canNodes*
 Add bidirectional connections from *canNodes* to q at *li* layer
 for every node $n \in$ *canNodes*
  Get the connections *nConnections* of node n at *li* layer
  If number of *nConnections* > *Max* (*Max*= 2 × *Max*, if *li*=0)
   Select *Max* closest nodes from *nConnections* as *nConnections_new*
   Update *nConnections* at layer *li* with *nConnections_new*
 ep = *W*

Fast image retrieval based on HNSW. Based on the established HNSW graphs using already registered images, overlapping image pairs of new fly-in image can be fast found during the HNSW updating procedure. As shown in Fig.5, the extracted global feature on new fly-in image is input into HNSW as an inserting node, via traversing the HNSW graphs from top to bottom, the corresponding existing nodes that are closest to the inserting node within each layer is retrieved, and then the final top-N result can be obtained from these existing nodes. In this work, the distance between nodes is estimated by the Euclidean distance of global feature and the method for searching each layer can be found at **Algorithm 2** (Search_layer).

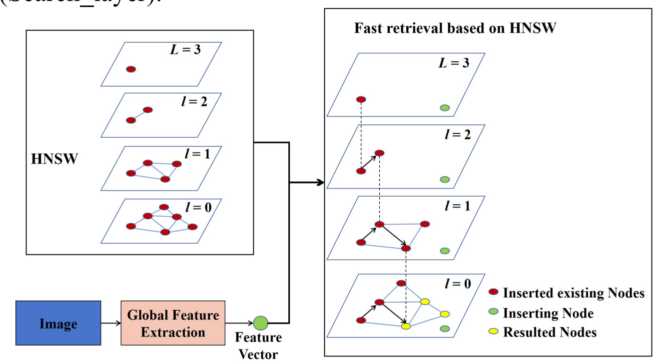


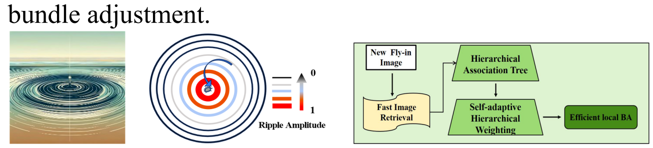
Fig. 5. Fast image retrieval based on HNSW.

Algorithm 2 Search_layer (q, *ep*, *efcon*, *li*)

Input inserting node q, enter point *ep*, layer number *li*, number of closest nodes to inserting node *efcon*.
Output *efcon* closest candidate nodes to q
1. **Initialization** v = *ep*: set of visited nodes, C = *ep*: set of candidates, W = *ep*: dynamic set of retrieved closest nodes.
2. **While** $|C| > 0$
 Extract closest node from C to q, denoted as c
 Obtain furthest node from W to q, denoted as f
 If distance(c, q) > distance(f, q)
  Break (All nodes in W are evaluated)
 for every c's neighbourhood *cn* at *li* layer (update C and W)
  if $cn \notin v$
   update v with $v \cup cn$
  if distance (cn, q)<distance(f, q) or $|W|$<*efcon*
  update C with $C \cup cn$ and W with $W \cup cn$
  if $|W|$>*efcon*
   remove furthest node from W to q
return W

C. Efficient Local Bundle Adjustment with Self-adaptive Hierarchical Weights

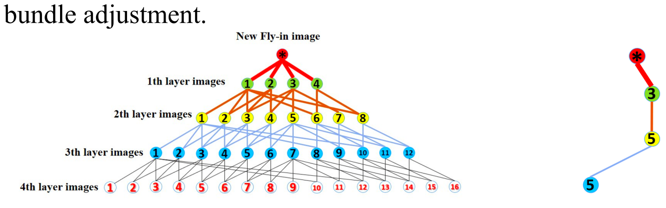
In order to achieve real-time performance for online reconstruction, it is crucial to employ a time efficient yet robust bundle adjustment. A common natural phenomenon is reminded that, throwing a stone into a peaceful lake, the resulted water wave always exhibits larger amplitude for the ripple that is closer to the stone as Fig.6(a) shows. So analogously, exemplified by Fig.6(b), throwing a new image into a well-solved photogrammetric block, images that have closer connection with this new image should be of higher influence, in another word, the uncertainty in a new fly-in image has a greater impact on closely related images than on those are further away. As illustrated in Fig.6(c), a novel efficient local bundle adjustment with self-adaptive hierarchical weights is used in our work: first, a hierarchical association tree is built based on image retrieval results (Section B), which reveals the association relationships between the new image and the previously registered images; then, we present a self-adaptive hierarchical weight for each locally associated image and employ them to perform a robust bundle adjustment.



(a) (b) (c)

Fig. 6. Hierarchical weighted local bundle adjustment inspired by water wave phenomenon.(a) Water wave; (b) Example indicating influence degree of new fly-in image (normalized within 0-1); (c) Workflow of our proposed bundle adjustment.

In order to achieve real-time performance for online reconstruction, it is crucial to employ a time efficient yet robust bundle adjustment. A common natural phenomenon is reminded that, throwing a stone into a peaceful lake, the resulted water wave always exhibits larger amplitude for the ripple that is closer to the stone as Fig.6(a) shows. So analogously, exemplified by Fig.6(b), throwing a new image into a well-solved photogrammetric block, images that have closer connection with this new image should be of higher influence, in another word, the uncertainty in a new fly-in image has a greater impact on closely related images than on those are further away. As illustrated in Fig.6(c), a novel efficient local bundle adjustment with self-adaptive hierarchical weights is used in our work: first, a hierarchical association tree is built based on image retrieval results (Section B), which reveals the association relationships between the new image and the previously registered images; then, we present a self-adaptive hierarchical weight for each locally associated image and employ them to perform a robust bundle adjustment.



(a) (b)

Fig. 7. Hierarchical association tree building and weighting. (a) Hierarchical association tree; (b) Example of shortest edges connecting new image to the 5-th image of 3-th layer, namely, $\varepsilon \to 3_5$.

1) Hierarchical association tree building and self-adaptive weighting.

Via the presented HNSW graphs-based efficient image retrieval method, it supposed to be very fast to find top-N similar images for new fly-in image. In addition, Fig.7(b) intuitively shows that images from different ripples (or hierarchical layers) should be with various influences caused by the addition of new fly-in image, which means they ought not to be treat equally in the optimization of bundle adjustment. Therefore, in this work, it makes sense to build a *Hierarchical association tree* for representing the correlations between already registered images and new image, whereby a reasonable weighting strategy can be then expected.

Images in the first ripple layer are the top-N similar images of the current new fly-in image, and the second ripple layers are generated by the top-N similar images of first ripple images, this process is repeated until a pre-setting depth Lh is reached. As shown in Fig.7, we give an illustration of a simplified 4-layer hierarchical association tree, where each bottom-layer image is a retrieved top-N image from the layer above it. All images included in hierarchical association tree are represented as I^hat. It is worth noting that the first ripple layer images exert the most significant influence caused by the new image, and the higher the ripple layer is, the less the corresponding images are affected. In addition, in the same layer, the weight of different images should be slightly different due to the various similarity degree with the new image. According to the above mentioned, a self-adaptive hierarchical weighting method for images in different ripples is proposed, as shown in (2):

$$p_{ij}=\begin{cases}1, & if\ i,j=*\\ 1/(\sum_{\varepsilon\to i_j}\frac{1}{s_\varepsilon})^{i-1}, & if\ i\neq L_h\\ \infty, & if\ i=L_h\end{cases} \qquad (2)$$

where i is the layer number, $*$ is the current new fly-in image, j indicates the image that is classified to i-th layer, S returns the similarity degree calculated by the Euclidean distance of two global features and $\varepsilon \to i_j$ contains set of shortest edges (weighted by similarity degrees, as Fig.7(b) implies) connecting new fly-in image and the j-th image of i-th layer.

2) Local bundle adjustment with self-adaptive hierarchical weights.

Bundle adjustment revisited. Let x be a vector of parameters for the camera and 3D points that need to be refined, and $f(x)=[f_1(x),\cdots,f_k(x)]$ denotes reprojection errors of each projection ray via collinearity equation. The goal of bundle adjustment is to find optimal x that minimize the reprojection error $f(x)$. Typically, this optimization problem is formulated as a non-linear least squares problem, with the total error defined as summed squares of residuals between the observed feature's position and the estimated 2D position from reprojection of the corresponding 3D point on the image, as shown in (3):

$$x^* = \arg\min_x E(x), \qquad (3)$$

where $E(x)=\sum_{i=1}^{k}\|f_i(x)\|^2$, k is the number of all projection rays.

Let x_t be the updated solution after t iterations, (3) can be approximated by Taylor expansion:

$$E(x)\approx E(x_t)+g^T(x-x_t)+\frac{1}{2}(x-x_t)^T H(x-x_t) \qquad (4)$$

where $g=dE/dx(x_t)$, and $H=d^2E/dx^2(x_t)$. To solve (4), Gaussian-Newton is well-known used to:

$$\frac{dE}{dx}=g+H(x-x_t)=0 \ \rightarrow\ x_{t+1}=x_t-H^{-1}g \qquad (5)$$

Let $J(x)$ be the Jacobian of $f(x)$, so $g=dE/dx(x_t)=2J^Tf$, and the Hessian matrix H can be approximately estimated by $2J^TJ$. In addition, in order to ensure that matrix H can be inverted, we can alternatively use the canonical Levenberg-Marquardt (LM) algorithm [35] to modify (5):

$$x_{t+1}-x_t=-(H+\lambda D^TD)^{-1}g. \qquad (6)$$

$D(x)$ is a non-negative diagonal matrix, often derived from the diagonal of the matrix $J(x)^TJ(x)$, while λ is damping factor, a non-negative parameter, that controls the strength of gradient descent.

An equivalent normal equation that solves the updated item δ is obtained as (7):

$$(J^TJ+\lambda D^TD)\delta=-J^Tf. \qquad (7)$$

The updated δ can be divided as $\delta=[\delta_c,\delta_p]$, in which δ_c is for camera parameters and δ_p is for 3D point parameters.

Then, assume $U = {J_c}^T J_c$, $V = {J_p}^T J_p$, $U_\lambda = U + \lambda {D_c}^T D_c$, $V_\lambda = V + \lambda {D_p}^T D_p$, and $W = {J_c}^T J_p$. As a result, (8) can be written as the block structured linear system:

$$\begin{bmatrix} U_\lambda & W \\ W^T & V_\lambda \end{bmatrix} \begin{bmatrix} \delta_c \\ \delta_p \end{bmatrix} = -\begin{bmatrix} {J_c}^T f \\ {J_p}^T f \end{bmatrix} \tag{8}$$

Based on Schur complement, a reduced normal equation system with only camera matrix can be obtained as:

$$\left(U_\lambda - W {V_\lambda}^{-1} W^T\right)\delta_c = -{J_c}^T f + W {V_\lambda}^{-1} {J_p}^T f \tag{9}$$

Then, camera parameter updates δ_c can be solved using (9) and δ_p is estimated by ${V_\lambda}^{-1}({J_p}^T f + W^T \delta_c)$.

BA integrated with self-adaptive hierarchical weights. To guarantee online SfM, based on the generated local photogrammetric block I^{hat} and the corresponding weights p_{ij}, we proposed a time and robustness efficient local bundle adjustment with self-adaptive hierarchical weights. In this work, (8) is improved and shown:

$$\begin{bmatrix} U_\lambda^{hat} P^{hat} & W_\lambda^{hat} \\ W_\lambda^{hat} P^{hat} & V_\lambda^{hat} \end{bmatrix} \begin{bmatrix} {\delta_c}^{hat} \\ {\delta_p}^{hat} \end{bmatrix} = -\begin{bmatrix} J_c^{hat^T} P^{hat} f^{hat} \\ J_p^{hat^T} f^{hat} \end{bmatrix} \tag{10}$$

where, only the local block BA with information regarding images I^{hat} are solved and P^{hat} is composed of corresponding weights p_{ij} in equation (2). The new reduced camera matrix

$$\left(U_\lambda^{hat} - W_\lambda^{hat} V_\lambda^{hat^{-1}} W_\lambda^{hat^T}\right) P^{hat} {\delta_c}^{hat} = -J_c^{hat^T} P^{hat} f^{hat} + W {V_\lambda}^{-1} J_p^{hat^T} f^{hat} \tag{11}$$

Via (11), we can faster and robustly calculate updated item ${\delta_c}^{hat}$, and 3D point ${\delta_p}^{hat}$ can be also efficiently obtained by $-V_\lambda^{hat^{-1}}(J_p^{hat^T} f^{hat} + W_\lambda^{hat^T} P^{hat} {\delta_c}^{hat})$.

Algorithm 3 3D similarity transformation

Input source submap $\boldsymbol{M_S}$, reference submap $\boldsymbol{M_R}$, minimum inlier ratio *min_ior*, maximum reprojection error *max_re*, RANSAC iterations *num_trials*
Output 3x4 3D similarity transformation matrix.
1. **Initialization** Obtain shared images $\boldsymbol{S_I}$, $\boldsymbol{R_I}$ and common 3D points $\boldsymbol{P_S}$ and $\boldsymbol{P_R}$ from $\boldsymbol{M_S}$ and $\boldsymbol{M_R}$, respectively. If $\boldsymbol{S_I}$ contains more than 3 images, go next step.
2. **Loop** for *num_trials*:
   2.1 Randomly select three shared images from $\boldsymbol{S_I}$ and $\boldsymbol{R_I}$, denoted as ***SrcRan*** and ***RefRan***, respectively.
   2.2 Estimate 3D similarity transformation matrix $\boldsymbol{T}$ based on ***SrcRan*** and ***SrcRan*** (Wang et al., 2019).
   2.3 Calculate the inlier ratio related to every *imageI* in $\boldsymbol{S_I}$ based on *imageR* in $\boldsymbol{R_I}$:
      2.3.1 For every 3D point $\boldsymbol{p}_s \in \boldsymbol{P_S}$ and observed by *imageI*, find matching 2D image point $\boldsymbol{x_R}$ from *imageR*.
      2.3.2 Transfer $\boldsymbol{p}_s$ into submap $\boldsymbol{M_R}$: $\boldsymbol{p}_s^{MR} = \boldsymbol{T}\boldsymbol{p}_s$.
      2.3.3 Reprojection error $re12 = \| \boldsymbol{x_R} - \text{reproj}\ (\boldsymbol{p}_s^{Mr})\|$, reproj(.) is the reprojection using orientation parameters of *imageR*
      2.3.4 Similarly, get reprojection error from *imageR* to *imageI*, $re21 = \| \boldsymbol{x_S} - \text{reproj}\ (\boldsymbol{p}_r^{Ms})\|$, and $\boldsymbol{p}_r^{Ms} = \boldsymbol{T}^{-1}\boldsymbol{p}_r$, $\boldsymbol{p}_r \in \boldsymbol{P_R}$
      if $re12 \le max\_re$ and $re21 \le max\_re$:
         For *imageI*, *num_inliers* = *num_inliers*+1.
      2.3.5 Inlier ratio of *imageI*: *num_inliers* / *num_common_points*, *num_common_points* is the point number of $\boldsymbol{P_S}$ or $\boldsymbol{P_R}$
   2.4 Obtain *numInlierImages* whose inlier ratios are higher than *min_ior*.
   2.5 Iterate steps 2.1 to 2.4, iteratively update the $\boldsymbol{T}$ with higher *numInlierImages*.
3. If the final highest *numInlierImages* is less than 3, then return **Fail**. Otherwise, return **Success** and final $\boldsymbol{T}$ related to highest *numInlierImages*.

D. Multiple Submaps Processing

The emergence of multiple submaps is a very common issue in online real-time solutions, such as [5], it can be generated when employing multiple agents or using one agent in a very arbitrary way. In this part, we introduce the most notable improvement of our on-the-fly SfMv2, which includes the processing of multiple submap and yields a complete reconstruction result. Consequently, the corresponding challenge is how to integrate different submaps and create a complete reconstruction result of the entire scenario. Fig.8 illustrates the general workflow of our multiple submap processing, the details steps are as follows:

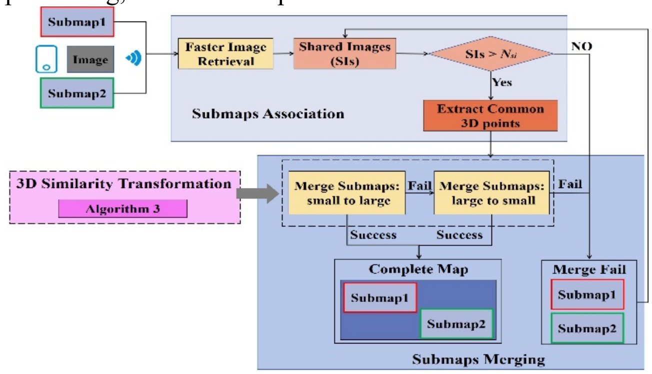


Fig. 8. Workflow of submaps processing.

Submaps Association. Given two independent submaps, when a new image flies in, our proposed fast image retrieval method is applied to find corresponding overlapping image pairs from already registered images in these two submaps, if online registrations are successfully performed on both submaps, a shared image is recorded. Once the number of shared images (*SIs*) is higher than a threshold (N_{si} = 3 in this paper), the common 3D points are found to run next step.

Submaps Merging. After obtaining commonly shared images and 3D points, the 3D similarity transformation (see **Algorithm 3** for more details) is applied to estimate the relative orientation between submaps, and one submap is transferred into the other one. Our on-the-fly SfMv2 firstly tries to merge the smaller submap (less registered images) into the larger one (in other words, the larger submap with more registered images is considered as a reference) to improve time efficiency and accuracy. If this initial fusion fails, an alternative is activated, i.e., the larger submap is merged into the smaller submap. As long as any of these two fusion attempts succeeds, the two submaps can be consolidated into one single yet complete submap. If both attempts fail, the two submaps remain disconnected, at the same time, the next new image flies in and additional shared images will be identified, after which another merge trial will be made.

By employing this method, two associated submaps merging can be fast initiated, which facilitates real-time submaps processing. For multiple submaps, pairwise fusion is applied in a recursive way until all submaps are processed.

IV. EXPERIMENTS

In this section comprehensive experiments are conducted to demonstrate the efficacy of our updated on-the-fly SfMv2, the primary goal of the experiments is to assess our new real time

SfM system as a whole and to support that you can *get better from what you capture*. In particular, extensive ablation studies are performed to validate the improvements on online image retrieval with HNSW, adaptive hierarchical weighting for local bundle adjustment, and the capability for dealing with multiple agents and multiple submaps. Then, numerical comparisons with mutual packages (Colmap and OpenMVG) and our previous on-the-fly SfM (v1) are reported to show the accuracy of our method regarding orientation result. All experiments are run on the machine with i9-12900K CPU and RTX3080 GPU.

A. Implementation Details

Multi-agents on-the-fly SfMv2 platform. Our previous v1 embeds a device (CAMFI 3.0[2]) to emit wireless WIFI signal for image transmission. In this work, to support the working mode of multiple agents and improve the flexibility of feasible sensors (such as cellphone, portable tablet PC), we instead designed and developed a convenient "client-server" transmission subsystem, as Fig.9 illustrates. More specifically, on the image capturing agents (as client), we developed an application based on android, which allows each captured image to be efficiently transmitted to the server via WLAN or 5G. On the server side, there is a corresponding receiving end to collect images sent by multiple agents and processing end to run our proposed updated SfMv2. In practice, one or more agents take images on different regions of the scene and transmit them in real-time to receiving end, and our on-the-fly SfMv2 (processing end) is recursively triggered to incrementally build and merge the sub-reconstructions in real time. In our tests, image captured by the agents can be successfully received by the server within approximately 3 seconds. More information and the code of this work can be accessed at https://rayshark0605.github.io/OnTheFly-SfM.github.io.

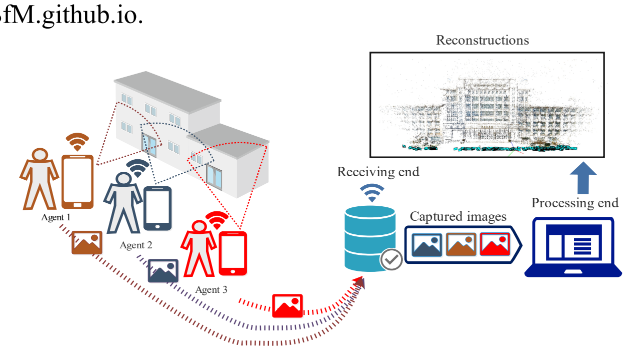


Fig. 9. Multi-agents on-the-fly SfMv2 platform.

Experimental datasets. As shown in Fig.10, seven datasets of various scenarios are tested to evaluate the proposed method. Four datasets (*YD*, *YX*, *XZL*, *JYYL*) are self-captured by multi-agents platform, in which *YD* (291 images) is UAV images captured without a specific regular flight, *YX* (349 images), *XZL*(226 images) and *JYYL*(356 images) are arbitrarily captured by using single, double and triple agents, respectively; three public datasets (*UniKirche*, *fr1_xyz* [3], *Alamo*) are simulated as sequential inputs to further demonstrate the corresponding superiority of proposed methods. *UniKirche* [31] contains 1455 unordered UAV and close-range images, based on the original stored order, several submaps should be expected which is just suitable to prove our corresponding capability for coping with multiple submaps. *fr1_xyz* is an ordered video dataset with 798 frames and is tested for comparing SfM results. *Alamo* [55] is crowdsourced dataset captured by different tourists and used for exemplifying the online image retrieval efficacy.

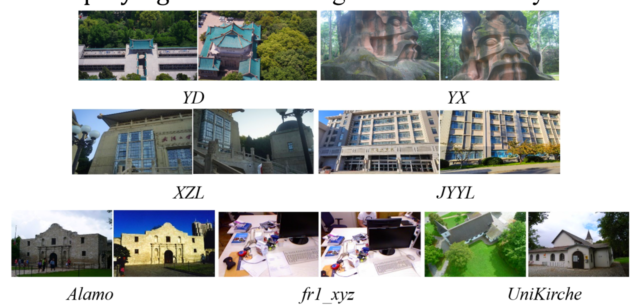


Fig. 10. Sample images of experimental datasets.

Running Parameters. In our experiments, there are some free parameters needed to be selected. For HNSW online image retrieval, we set *max_elements* to be 10000 which indicates that the corresponding HNSW graph can store up to 10,000 data points (it can be changed according to specific task), *ef_construction* = 200 (each element's dynamic candidate list contains up to 200 elements during graph construction) and M = 16 (each data point in the graph is connected to at most 16 other points). For online sub-reconstruction, each newly fly-in image selects the Top-30 most similar images for subsequent image matching and geometric verification. In the hierarchical association tree and efficient local bundle adjustment, we set the maximum number of levels L_h be 4 with Top-8 most similar images to ensure that a sufficient number of images are involved in the optimization of bundle adjustment, while not overly decrease the time efficiency due to an excessive BA block. To merge connected submaps, the *Nsi* is set to be 3.

B. Performance of the Proposed Faster Image Retrieval Based on HNSW

In this section, four datasets (*YD*, *UniKirche*, *XZL*, and *Alamo*) are employed to evaluate our proposed faster image retrieval method, and the performance of retrieval precision and time efficiency are compared with two state-of-the-art methods, i.e., exhaustive retrieval that compares all already registered images and the newly fly-in image and vocabulary tree with the same settings of our previous on-the-fly SfM (v1), note that the same learning-based global features are used. For this experiment, images of *UniKirche* and *Alamo* are simulated to be captured by the inherent order as they are stored. To obtain referenced overlapping images for each new image, the corresponding image pairs that can pass the

[2] More details of CAMFI3.0 can be found at https://www.cam-fi.com/en/index.html

[3] TUM dataset: https://cvg.cit.tum.de/data/datasets/rgbd-dataset

conventional two-view geometry verification[4] are cast as referenced results. Finally, sorting by descending order of inlier match count provided the Ground Truth for the Top-30 images at the time of their flown-in.

Tab. I compares the precision of the three methods when retrieving the Top-30 matchable candidate images. The results indicate that HNSW can find more true positives than vocabulary tree method and is nearly on the pair with exhaustive retrieval method, especially for crowdsourced images HNSW is able to achieve the same precision as exhaustive retrieval. This can be explained by the fact that HNSW is composed of hierarchical graphs that connect most neighboring nodes, which means nearly all potential connected images are traversed, whereas, vocabulary tree might result in ambiguous results from the unevenly divided feature space.

TABLE I

COMPARISON OF RETRIEVAL PRECISION FROM VARIOUS METHODS' TOP-30

Method	*YD*	*UniKirche*	*XZL*	*Alamo*
Exhaustive match	**87.14**	**71.31**	**64.71**	**84.10**
Vocabulary tree	71.04	55.13	50.13	35.04
HNSW	86.54	71.31	62.54	**84.10**

Considering the real-time performance, the time efficiency is listed in Tab. II and a qualitative time cost result of *Alamo* is shown in Fig.11, in which the trend curves of consuming time for the three methods are fitted as the fly-in image increases. It can be observed that the averaging cost time of HNSW is the lowest, the vocabulary tree ranks in the middle and exhaustive retrieval method perform the slowest, this confirms that our proposed method is a faster image retrieval method. In addition, from Fig.11, the cost time of exhaustive retrieval linearly increases as number of involved images grows and this can be expected due to more images are needed to be compared, whereas, both HNSW and vocabulary tree tend to be stable and with slight fluctuations when more and more new images fly in, which is primarily attributed to the efficient indexing structure and search strategies. HNSW applies navigational strategies across different layers and ensures a relatively consistent search time even with a significant increase in data volume, as it can quickly locate in the higher-level graph and then perform precise searches at lower-level graphs. Vocabulary tree traverses the new fly-in images by comparing it with the cluster centers, which results in constant computations for a pre-constructed vocabulary tree. However, fluctuations in retrieval time might happen due to extra refinement when some ambiguous candidates are found.

In general, the proposed HNSW-based image retrieval method can improve our previous on-the-fly SfM by providing more accurate overlapping images yet in a more time efficient way.

TABLE II

COMPARISON OF THE AVERAGED CONSUMING TIME (IN MS). BESTS ARE HIGHLIGHTED IN BOLD

Method	*YD*	*UniKirche*	*XZL*	*Alamo*
Exhaustive match	634	1098	761	1439
Vocabulary tree	604	402	750	439
HNSW	**581**	**360**	**653**	**364**

[4] After extracting and matching SIFT features, image pairs with more than 50 inlier correspondences that conforms two-view geometry are considered referenced overlapping image pairs.

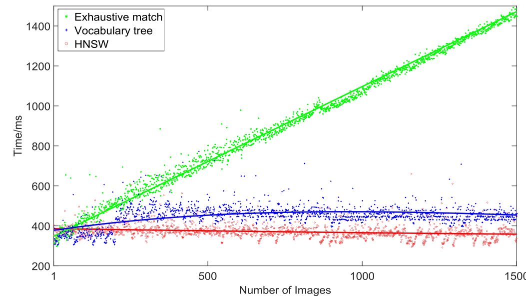


Fig. 11. Cost time on *Alamo* with various retrieval methods (the first 1500 images are shown).

C. Performance of Adaptive Weighting Local BBundle Adjustment

To validate the effectiveness of the proposed self-adaptive hierarchical weighting local BA, we conducted experiments on four datasets: *YD*, *XZL*, *YX*, and *UniKirche*. We recorded three criteria: the averaged mean reprojection error (AMRE) and the averaged adjustment time (AT) of all local BA for optimizing new fly-in images, and the mean final reprojection error (MFRE) of the last global adjustment. The conducted experiments include our previous on-the-fly SfM (v1), the method presented in this paper (v2), and Colmap. Notably, for a fair comparison, the AT for Colmap does not include the original iterative global BA. The relevant results are shown in Tab. III.

TABLE III

COMPARISON OF VARIOUS BA METHODS

Method	*YD*			*XZL*			*YX*			*UniKirche*		
	AMRE	MFRE	AT (in ms)	AMRE	MFRE	AT (in ms)	AMRE	MFRE	AT (in ms)	AMRE	MFRE	AT (in ms)
v1	0.70	1.16	196	0.81	0.68	390	1.10	0.58	**302**	0.84	0.66	295
v2	**0.46**	**1.07**	207	**0.68**	**0.62**	399	**0.56**	**0.53**	352	**0.43**	**0.39**	348
Colmap	0.54	1.13	**170**	0.79	0.65	**368**	0.63	0.56	329	0.52	0.42	**254**

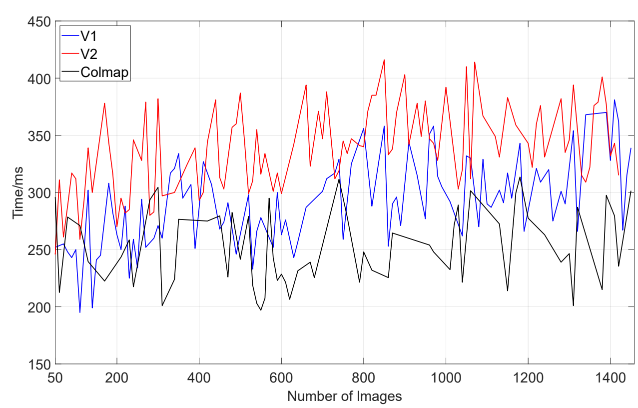


Fig. 12. Cost time of various bundle adjustment methods on *UniKirche*. The horizontal axis demotes the numbers of new fly-in images.

It can be seen that the usage of our adaptive hierarchical weighting strategy for local BA can help to achieve the best AMRE and MFRE, but there is a slight increasing in bundle adjustment time AT. Fig.12 shows that v2 generally takes more time than v1 does (with just negligible magnitude if considering AT itself) and Colmap is basically the fastest solution for local BA. This could be expected and resulted from the employed hierarchical weighting strategy. First, the built hierarchical association tree typically contributes to a larger photogrammetric block than the Colmap local BA that typically selects fix number of enrolled images does, thus, Colmap can run local BA more efficiently than v1 and v2 do. Second, instead of vocabulary tree that is integrated with v1, v2 applies a better online image retrieval method to construct hierarchical association tree, and this could lead to more already registered images to be accurately retrieved, thereby more images are included in local BA of v2 (as Fig.13 implies). Third, this work adopts a self-adaptive hierarchical weighting mechanism, and reasonably estimates the weight between new fly-in image and already registered images via the similarity degree values, therefore, the best performance of BA is always obtained by v2.

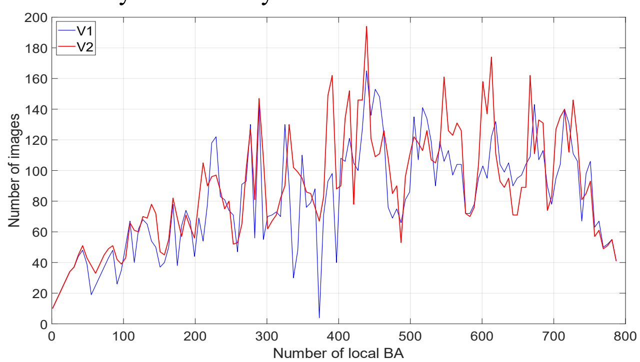


Fig. 13. The number of images participate in each local bundle adjustment.

D. Capability for Multiple Agents and Sub-reconstructions

In this part, to demonstrate the capability of coping with multiple agents and sub-reconstructions, we select four challenging datasets (*XZL*, *JYYL, UniKirche* and *YD*) to illustrate two possible scenarios encountered in practice: first, multiple agents for multiple sub-reconstructions. Agents initially take images independently in various disconnected submaps, then continue to capture the overlapping areas among these submaps for merge multiple submaps into a complete map. In this case, *XZL* and *JYYL* with two and three agents are used; Second, single agent with multiple sub-reconstructions. The single employed agent takes images in a totally arbitrary manner, which may result in fragmental submaps. *UniKirche* and *YD* captured by just one agent are tested in this scenario.

Intuitively, Fig.14 to Fig.17 shows the reconstruction results of the selected datasets using the updated on-the-fly SfMv2 and our previous on-the-fly SfMv1, it can be obviously seen that a better and complete reconstruction map is generated by v2, whereas v1 produces many fragmental sub-reconstructions. Quantitative information is presented in Tab. IV, where Tab. IV-I lists the number of sub-reconstructions and Tab. IV-II compiles the number of registered images and the number of 3D points in each reconstruction.

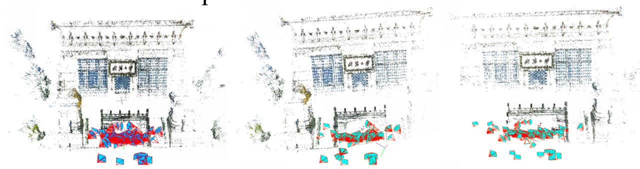


Fig. 14. Reconstruction results of *XZL*. Left: complete reconstruction by v2. Middle and Right: two sub-reconstructions by v1.

Fig. 15. Reconstruction results of *UniKirche*. Left: complete reconstruction by v2. Middle and right: two sample sub-reconstructions by v1.

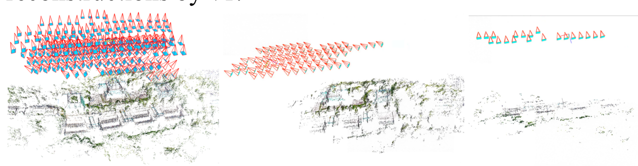


Fig. 16. Reconstruction results of *YD*. Left: complete reconstruction by v2. Middle and right: two sub-reconstructions by v1.

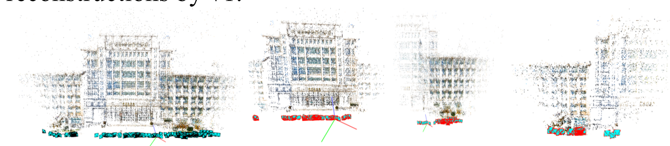


Fig. 17. Reconstruction results of *JYYL*. Left: complete reconstruction by v2. Remaining three on the right: three sub-reconstructions within v1.

The above results explicitly demonstrate superiority of our updated v2 for dealing with multiple agents and sub-reconstructions. Our previous on-the-fly SfM is basically not able to take care of multiple agents, even within same area, if capturing procedure is too arbitrary and the overlapping area between adjacent images is too low, it is very likely to generate multiple submaps, leading to multiple reconstructions in the results. Nevertheless, our method can handle most of these challenges by recursively attempting to merge submaps for multiple agents and sub-reconstructions. This success can be attributed to the 3D similarity transformations and fusion algorithm between submaps.

TABLE IV-I
THE NUMBER OF SUB-RECONSTRUCTIONS IN THE FINAL RESULTS

Datasets	v2	v1
XZL	1	2
UniKirche	1	13
YD	1	5
JYYL	1	3

TABLE IV-II
THE NUMBER OF IMAGES AND 3D POINTS IN V1 AND V2

Datasets		v2	v1			
			Sub-reconstruction 1	Sub-reconstruction 2	Sub-reconstruction 3	……
XZL	images	226	172	155	——	——
(226 images)	points	153699	78880	65121	——	——
UniKirche	images	1454	186	35	105	……
(1455 images)	points	586687	96048	50373	38762	……
YD	images	291	226	26	20	……
(291 images)	points	129321	58409	2295	4076	……
JYYL	images	354	216	153	200	——
(354 images)	points	179327	94906	66288	89960	——

E. Comparison with the State-of-the-art SfM Methods

To further investigate how deep the gap between our v2 and the mutual SfM methods is, two popular offline SfM packages Colmap and OpenMVG are compared as references, as well as our previous on-the-fly SfM (v1). After the final reconstruction completes, the evaluation is performed on three criteria: the mean reprojection error (MRE), the mean tracking length (MTL), and the mean rotation discrepancy (MRD) taking results of Colmap and OpenMVG as reference. The results of corresponding experiments are shown in Tab. V and Tab. VI.

TABLE V
COMPARISON WITH COLMAP AND OUR PREVIOUS ON-THE-FLY SFM(V1)

Datasets	MRE			MTL			MRD (in degree)	
	v2	v1	Colmap	v2	v1	Colmap	v2-Colmap	v1-Colmap
UniKirche	1.09	1.97	**1.07**	5.13	4.21	**5.87**	**0.48**	0.68
YX	**1.17**	2.17	1.24	10.24	9.38	**12.04**	**0.35**	0.63
YD	**1.14**	2.28	1.18	3.75	3.11	**4.33**	**0.44**	0.54
XZL	**1.42**	2.31	1.43	7.40	5.47	**7.55**	**0.36**	0.92
JYYL	**1.39**	2.44	1.40	6.53	4.65	**6.46**	**0.35**	0.54
fr1_xyz	**1.05**	1.87	1.15	**46.52**	32.56	45.17	**0.33**	0.53

TABLE VI
COMPARISON WITH OPENMVG AND OUR PREVIOUS ON-THE-FLY SFM(V1)

Datasets	MRE			MTL			MRD (in degree)	
	v2	v1	OpenMVG	v2	v1	OpenMVG	v2-OpenMVG	v1-OpenMVG
UniKirche	1.09	1.97	**0.79**	**5.13**	4.21	4.00	**0.73**	0.94
YX	1.17	2.17	**0.50**	10.24	9.38	**14.00**	**0.32**	0.66
YD	1.14	2.28	**0.87**	3.75	3.11	**5.00**	**1.07**	1.17
XZL	1.42	2.31	**0.82**	**7.40**	5.47	6.00	**0.45**	1.00
JYYL	1.39	2.44	**1.10**	**6.53**	4.65	4.00	**0.71**	0.76
fr1_xyz	1.05	1.87	**0.86**	**46.52**	32.56	12.00	**0.73**	0.93

In general, compared to Colmap, our work can achieve better MRE on most datasets. This is primarily due to our consideration of additional indirect corresponding images in the local bundle adjustment, as discussed before (**Section III.C** and **Section IV.C**), this slightly reduces the time efficiency of local BA but results in better interior precision, i.e., smaller MRE. On the other hand, OpenMVG outperforms Colmap, v1 and v2, this is because the inherent BA of OpenMVG sets two tough constraints to directly eliminate ineligible observations: minimum intersection angle must be larger than 2 degrees and the reprojection residual must be smaller than 2 pixels, which also explains that some MTL values of OpenMVG are smaller (in particular for *fr1_xyz*, which contain sequential frames with very short baselines and tinny intersection angles). The MTL of v2 is closer to Colmap than v1, this might be due to the high retrieval precision of the proposed HNSW-based image retrieval method that can lead to more matches, consequently, more 3D points are triangulated. In regard to the orientation precision, taking the mutual packages Colmap and OpenMVG as reference, our v2 always yields more precise rotations than v1 does, indicating better reconstruction results are obtained from the captured images.

V. CONCLUSION

In this work, we have extended our previous on-the-fly SfM [63] with several remarkable updates to "*get better from what you capture*". More specifically, three significant improvements are made: first, a faster online image retrieval method based on HNSW is proposed to get more accurate overlapping images; second, local bundle adjustment is improved by integrating with a hierarchical self-adaptive weighting strategy; third, making feasibility for dealing with multiple agents and submaps.

As our comprehensive experimental results demonstrate, our updated on-the-fly SfMv2 is capable to run online SfM using multiple agents and can "*get better from what you capture*", in which "*better*" means more complete and more accurate reconstruction result can be obtained. In the future, with the goal of developing a relatively complete online photogrammetric processing system, we would like to further explore and update our v2 along the direction of generating dense point cloud and surface mesh in real time.

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